

Complex Number 1

1. By equating real and imaginary parts, find the real values of x and y given that:
 - a. $(x + yi)(2 - 3i) = -13i$
 - b. $(1 + 4i)(x + yi) = 6 + 7i$
 - c. $(1 + i)x + (2 - 3i)y = 10$
 - d. $x(1 + 2i) + y(2 - i) = 4 + 5i$
 - e. $\frac{x}{2+i} + \frac{y}{2+3i} = 4 + i$
2. Let $z = a + ib$, where a and b are real and non-zero. Prove that:
 - a. $z + \bar{z}$ is real,
 - b. $z - \bar{z}$ is imaginary,
 - c. $z^2 + (\bar{z})^2$ is real,
 - d. $z\bar{z}$ is real and positive.
3. Let $z = a + bi$, where a and b are real. If $\frac{z}{z-i}$ is real, show that z is imaginary.
4. Prove that if $z^2 = (\bar{z})^2$ then z is either purely real or imaginary but not complex.
5. If $z = x + iy$, where x and y are real, express in the form $a + ib$, where a and b are written in terms of x and y :
 - a. z^{-1}
 - b. z^{-2}
 - c. $\frac{z-1}{z+1}$
6. If both $z + w$ and zw are real, prove that either $z = \bar{w}$ or $\text{Im}(z) = \text{Im}(w) = 0$.
7. Solve for z :
 - a. $z^2 - z + (1 + i) = 0$
 - b. $z^2 + 3z + (4 + 6i) = 0$
 - c. $z^2 - 6z + (9 - 2i) = 0$
 - d. $(1 + i)z^2 + z - 5 = 0$
 - e. $z^2 + (2 + i)z - 13(1 - i) = 0$
 - f. $iz^2 - 2(1 + i)z + 10 = 0$
8. Find the two complex number z satisfying $z\bar{z} = 5$ and $\frac{z}{\bar{z}} = \frac{1}{5}(3 + 4i)$
9. Consider the quadratic equation $ax^2 + bx + c = 0$, where a, b and c are real and $b^2 - 4ac < 0$. Suppose that w is one of the complex roots of the equation.
 - a. Explain why $aw^2 + bw + c = 0$
 - b. By taking the conjugate of both sides of the result in part a., and using the properties of conjugates, show that $a(\bar{w})^2 + b(\bar{w}) + c = 0$.
 - c. What have you just proved about the two complex roots of the equation?
10. Find the discriminant and its square roots, and hence solve:
 - a. $z^2 + (4 + 2i)z + (1 + 2i) = 0$
 - b. $z^2 - 2(1 + i)z + (2 + 6i) = 0$
 - c. $z^2 + 2(1 - i\sqrt{3})z + 2 + 2i\sqrt{3} = 0$
 - d. $z^2 + (1 - i)z + (i - 1) = 0$
11. The points P and Q represent the complex numbers z and w on an Argand diagram, where $w = \frac{3z+2}{z-1}$ and P moves on the circle $|z| = 1$. By writing an expression for z in terms of w , or otherwise, sketch the locus of w .

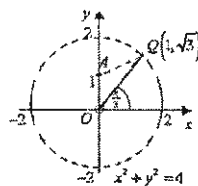
12. a. Sketch and shade on an Argand diagram the region R in which $|z - 4i| \leq 3$ and $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$ hold simultaneously.
- b. Find the value of $\arg(z + 1)$ at the point in the region R where $\arg(z + 1)$ is a minimum. Answer to the nearest degree.
13. Use the following identity to answer the following questions: $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$
- a. Solve $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$
- b. Hence show that
- i. $\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + 4\sqrt{3} = \tan \frac{5\pi}{24} + \tan \frac{11\pi}{24}$
- ii. $\tan \frac{\pi}{24} \tan \frac{5\pi}{24} = \cot \frac{7\pi}{24} \cot \frac{11\pi}{24}$
- c. Find the polynomial of least degree that has zeros $\left(\cot \frac{\pi}{24}\right)^2, \left(\cot \frac{7\pi}{24}\right)^2, \left(\cot \frac{13\pi}{24}\right)^2, \left(\cot \frac{19\pi}{24}\right)^2$
14. z is a complex number such that $\arg z = \frac{\pi}{3}$ and $|z| \leq 2$
- a. Show the locus of the point P representing z in the argand diagram.
- b. Find the possible values of the principal argument of $z - i$ for z on this locus.
15. Let $w = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$
- a. Find all the complex roots of the equation $z^{10} - 1 = 0$ by writing them in terms of w and k , while k is a positive integer.
- b. Prove that $1 + w + w^2 + \dots + w^9 = 0$
- c. The quadratic equation $x^2 + bx + c = 0$, where b and c are real, has the root $w + w^4$. Find the other root in terms of w .
- d. Find b and express c in terms of $\sin \frac{\pi}{5}$

Answer:

1. a. $x = 3$ and $y = -2$ b. $x = 2$ and $y = -1$ c. $x = 6$ and $y = 2$ d. $x = \frac{14}{5}$ and $y = \frac{3}{5}$ e. $x = \frac{35}{2}$ and $y = -\frac{39}{2}$
5. a. $\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$ b. $\frac{x^2-y^2}{(x^2+y^2)^2} - \frac{2xy}{(x^2+y^2)^2}i$ c. $\frac{x^2+y^2-1}{(x+1)^2+y^2} + \frac{2y}{(x+1)^2+y^2}i$
7. a. $z = 1 - i$ or i b. $z = -3 + 2i$ or $-2i$ c. $z = 4 + i$ or $2 - i$ d. $z = -2 + i$ or $\frac{1}{2}(3 - i)$ e. $z = -5 + i$ or $3 - 2i$ f. $z = 3 + i$ or $-1 - 3i$
8. $z = \pm(2 + i)$
9. $w = w$ satisfies the equation. c. They are complex conjugates
10. a. $-2 - i \pm (\sqrt{\sqrt{2} + 1} + i\sqrt{\sqrt{2} - 1})$ b. $1 + i \pm (\sqrt{\sqrt{5} - 1} - i\sqrt{\sqrt{5} + 1})$ c. $-1 + i\sqrt{3} \pm (\sqrt{2} - i\sqrt{6})$ d. $\frac{1}{2}(-(1 - i) \pm (\sqrt{\sqrt{13} + 2} - i\sqrt{\sqrt{13} - 2}))$
11. $|w + 2| = |w - 3|$, locus is $x = \frac{1}{2}$



12. a. $\dots$ b. $\arg(z + 1) = 29^\circ$
13. a. $x = \tan \frac{\pi}{24}, \tan \frac{7\pi}{24}, \tan \frac{13\pi}{24}, \tan \frac{19\pi}{24}$ c. polynomial is $x^4 - 60x^3 + 134x^2 - 60x + 1 = 0$



14. a. $-\frac{\pi}{2} < \arg(z - i) \leq \tan^{-1}(\sqrt{3} - 1)$
15. a. $w, w^2, w^3, w^4, w^6, w^7, w^8, w^9$ c. the other root is $-2i \sin \frac{\pi}{5}$ d. $b = 0$ and $c = 4 \sin^2 \frac{\pi}{5}$